Lab Exercise 7- Install Minikube on Linux (Ubuntu /MacOS/Windows)

Install Minikube on Linux (Ubuntu)

```
sudo apt install -y docker.io # Ubuntu
```

```
sudo systemctl start docker

sudo systemctl enable docker

sudo usermod -aG docker $USER

newgrp docker
```

Step 2: Install kubectl

```
curl -LO https://storage.googleapis.com/kubernetes-release/release/$(curl -s

https://storage.googleapis.com/kubernetes-

release/release/stable.txt)/bin/linux/amd64/kubectl

chmod +x kubectl

sudo mv kubectl /usr/local/bin/
```

Verify:

```
kubectl version --client
```

Step 3: Install Minikube

```
curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
chmod +x minikube-linux-amd64
sudo mv minikube-linux-amd64 /usr/local/bin/minikube
```

Verify:

```
minikube version
```

Step 4: Start Minikube

```
minikube start --driver=docker
```

Check status:

```
minikube status
```

Install Minikube on Windows

Prerequisites

- Docker Desktop installed
- Enable WSL2

Install Minikube

```
choco install minikube -y
```

OR download exe:

```
https://github.com/kubernetes/minikube/releases/latest
```

Start:

```
minikube start
```

Install Minikube on macOS

```
brew install minikube

brew install kubectl
```

Start:

```
minikube start
```

Verify Kubernetes Cluster

```
kubectl get nodes
```

```
Expected output:

NAME       STATUS   ROLES          AGE   VERSION

minikube   Ready    control-plane   xx    v1.xx
```

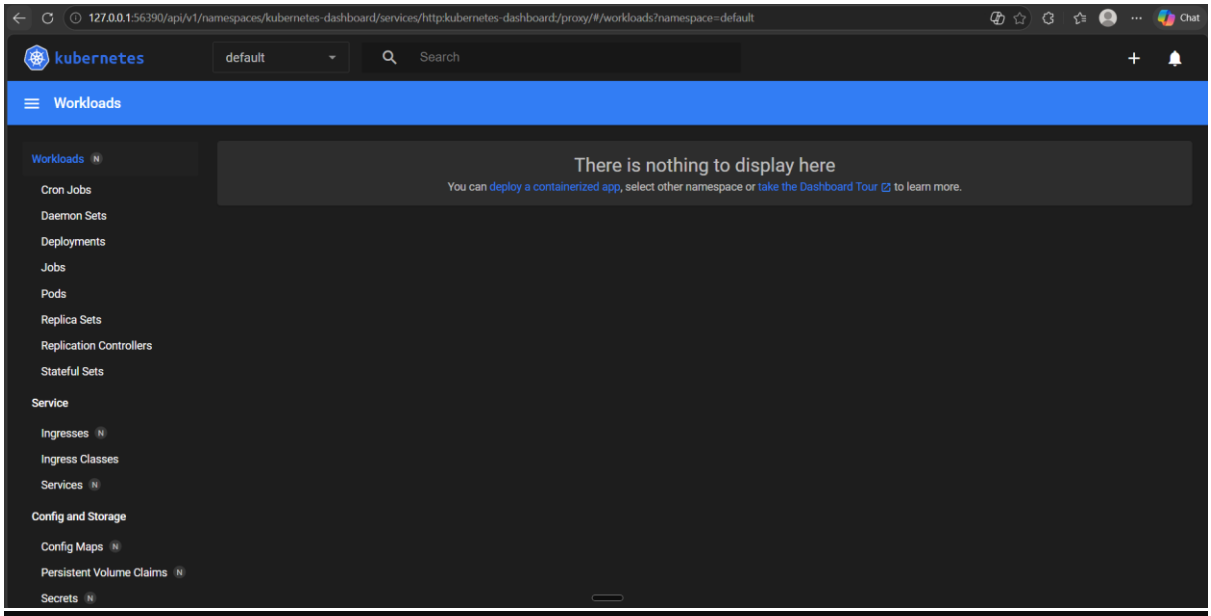
Useful Minikube Commands (Lab Ready)

Command	**Purpose**
minikube dashboard	Open K8s UI
minikube stop	Stop cluster
minikube delete	Delete cluster

Command	Purpose
minikube ssh	Access node
kubectl get pods -A	View all pods

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NK>minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0

C:\Users\NK>kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1

C:\Users\NK>minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 11 Home 25H2
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  234.65 MiB / 272.45 MiB  86.12% 12.26 Mi
```

```
* minikube v1.38.1 on Microsoft Windows 11 Home 25H2
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 13.80 M
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 14.39 M
* Creating docker container (CPUs=2, Memory=4000MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

C:\Users\NK>minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured


C:\Users\NK>kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   18s   v1.35.1

C:\Users\NK>minikube dashboard
* Enabling dashboard ...
  - Using image docker.io/kubernetesui/dashboard:v2.7.0
  - Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
* Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

* Verifying dashboard health ...
* Launching proxy ...
* Verifying proxy health ...
```




```
        minikube addons enable metrics-server

* Verifying dashboard health ...
* Launching proxy ...
* Verifying proxy health ...
* Opening http://127.0.0.1:56390/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default bro
^C
C:\Users\NK>kubectl get pods -A
NAMESPACE              NAME                                         READY   STATUS    RESTARTS       AGE
kube-system            coredns-7d764666f9-4xqw2                     1/1     Running   0              2m24s
kube-system            etcd-minikube                                1/1     Running   0              2m31s
kube-system            kube-apiserver-minikube                      1/1     Running   0              2m31s
kube-system            kube-controller-manager-minikube             1/1     Running   0              2m31s
kube-system            kube-proxy-jp8zt                             1/1     Running   0              2m24s
kube-system            kube-scheduler-minikube                      1/1     Running   0              2m31s
kube-system            storage-provisioner                          1/1     Running   1 (2m2s ago)   2m28s
kubernetes-dashboard   dashboard-metrics-scraper-5565989548-8h2k2   1/1     Running   0              2m5s
kubernetes-dashboard   kubernetes-dashboard-b84665fb8-jp7bc         1/1     Running   0              2m5s

C:\Users\NK>minikube stop
* Stopping node "minikube"  ...
* Powering off "minikube" via SSH ...
* 1 node stopped.

C:\Users\NK>minikube delete
* Deleting "minikube" in docker ...
* Deleting container "minikube" ...
* Removing C:\Users\NK\.minikube\machines\minikube ...
* Removed all traces of the "minikube" cluster.
```